

Bilal Ahmed

Senior AI Engineer

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Senior AI Engineer with 6+ years of experience designing and deploying production-grade AI systems and scalable backend architectures. Specialized in LLM-powered applications, including RAG pipelines, prompt engineering, and vector search. Experienced in building high-performance AI services using Python (FastAPI, Django) and Node.js, and deploying cloud-native systems on AWS and Kubernetes. Proven track record of integrating OpenAI, Hugging Face, and multimodal models into enterprise applications to automate workflows, improve user engagement, and deliver measurable performance gains.

Technical Skills

- **Programming Languages**

Python, JavaScript, TypeScript

- **AI / Machine Learning**

LLMs, RAG, Prompt Engineering, LangChain, LangGraph

LlamaIndex Transformers, RLHF, Fine-Tuning, NLP, PyTorch,

TensorFlow, NLP Multi-Agent Systems, Tool Calling, Function Calling

- **Backend Development**

FastAPI, Node.js, Django, REST APIs, GraphQL

Microservices Architecture, WebSockets, Event-Driven Systems

- **Frontend Development**

React, Next.js, TypeScript, TailwindCSS, Angular, Vue.js

- **Data & ML Infrastructure**

ML Pipelines, MLflow, Vector Databases, Embeddings, Semantic Search

- **Cloud & DevOps**

AWS (EC2, S3, RDS, ECS, ECR), Docker, Kubernetes

CI/CD, GitHub Actions, Jenkins, Apache Kafka

- **Automation & AI Workflows**

n8n, Zapier, Workflow Automation

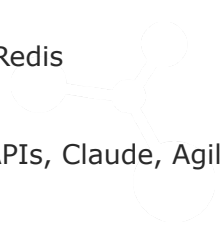
AI Orchestration, Event-Based Systems

- **Databases**

PostgreSQL, MongoDB, MySQL, Redis

- **Tools**

Git, GitHub, Bitbucket, OpenAI APIs, Claude, Agile/Scrum



Professional Experience

Senior AI Engineer

The Codest | Nov 2023 – Present

- Designed and deployed LLM-powered applications using Retrieval-Augmented Generation (RAG) to improve response accuracy and contextual understanding.
- Built scalable AI microservices using Python and FastAPI to support real-time inference and high-throughput systems.
- Integrated OpenAI, Hugging Face, and Llama models into enterprise platforms for conversational AI and workflow automation.
- Implemented vector search systems using Pinecone and ChromaDB to enable semantic search and enterprise knowledge retrieval.
- Collaborated with product and engineering teams to deliver AI-driven automation workflows that increased operational efficiency by 30%.

AI Engineer

Northcoders | Mar 2022 - Oct 2023

- Built full-stack applications integrating AI-powered backend systems with modern frontend frameworks, improving scalability and user experience.
- Developed LLM-driven features including intelligent search, conversational assistants, and recommendation systems.
- Designed and maintained scalable APIs using Python (FastAPI) and Node.js within microservices architecture.
- Implemented RAG pipelines to connect LLMs with enterprise data, enabling context-aware responses.
- Increased user engagement by ~30% through AI-enhanced search and recommendation capabilities.

Full Stack Developer

Innovify | Jul 2020 - Feb 2022

- Developed full-stack web applications using Python (FastAPI) and React/TypeScript for scalable and maintainable systems.
- Built high-performance REST APIs supporting real-time data processing and complex business workflows.
- Architected backend services capable of handling high-volume data pipelines and concurrent user requests.
- Optimized PostgreSQL database queries and indexing strategies, reducing query execution time by 25%.
- Deployed and maintained applications on AWS infrastructure to ensure reliable and scalable cloud services.

Projects

Amelia – Enterprise Conversational AI Platform

Technologies: Python, FastAPI, RAG, LangGraph, Vector Databases, AWS

- Built conversational AI system for automated customer support and enterprise workflows.
- Implemented RAG-based knowledge retrieval for accurate context-aware responses.
- Developed multi-agent orchestration pipelines for automation and decision support.

Design.ai – AI Content Generation Platform

Technologies: Python, Node.js, LangChain, LLM APIs, AWS

- Developed an AI platform capable of generating text, images and videos.
- Integrated multiple generative AI models for multimodal content creation.
- Built scalable backend services and cloud-based deployment architecture.

Sabermine – AI Document Processing System

Technologies: Python, FastAPI, OCR, NLP, AWS

- Built an AI system for automated document parsing and structured data extraction.
- Processed PDFs and scanned documents using NLP pipelines and OCR.
- Designed scalable processing pipelines for high-volume document workflows.

Myaigi – Multi-Agent AI System

Technologies: Node.js, JavaScript, LangChain, LangGraph, LLMs, RAG

- Developed multi-agent AI orchestration system for complex workflows.
- Built modular agent architecture for task-specific AI execution.
- Implemented reasoning pipelines using LLM-based decision systems.

SeekSocial – Social Learning Platform

Technologies: React, Node.js, PostgreSQL

- Built a collaborative learning platform, increasing user engagement by 25%.
- Implemented scalable backend services for real-time user interaction.
- Optimized APIs and queries, reducing response time by 20%.

Certifications

- ChatGPT Prompt Engineering for Developers – DeepLearning.AI
- LLM Applications Development with LangChain and OpenAI
- AWS Certified Developer – Associate

EDUCATION

Bachelor's degree (Computer Science)

University of Leeds, Leeds, United Kingdom